14th ASIAN AEROSOL CONFERENCE (AAC 2025)
1st - 4th December 2025, Mumbai, India
Theme - Aerosols: Shaping Air Quality and Climate Action

DEVELOPMENT AND DEPLOYMENT OF A LOW-COST IoT-BASED ENVIRONMENTAL QUALITY MONITORING SYSTEM (EQMS) WITH SPATIO-TEMPORAL PREDICTIVE MODELING

-Manuscript ID AACI139-

PRINCE VATS[1], ALOK K. THAKUR[2], SAMEER PATEL[1,3,4*]

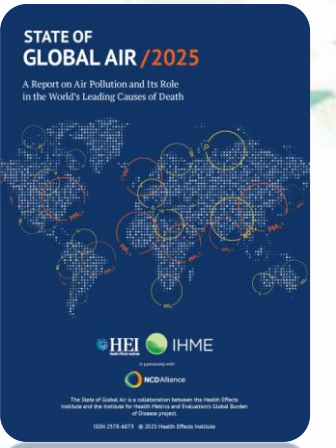
1. Department of Civil Engineering, IIT Gandhinagar
2. Department of Earth Sciences, IIT Gandhinagar
3. Department of Chemical Engineering, IIT Gandhinagar
4. Dr. Kiran C Patel Centre for Sustainable Development, IIT Gandhinagar

sameer.patel@iitgn.ac.in

ENERGY, ENVIRONMENT, AND EXPOSURE LABORATORY
INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

Motivation:

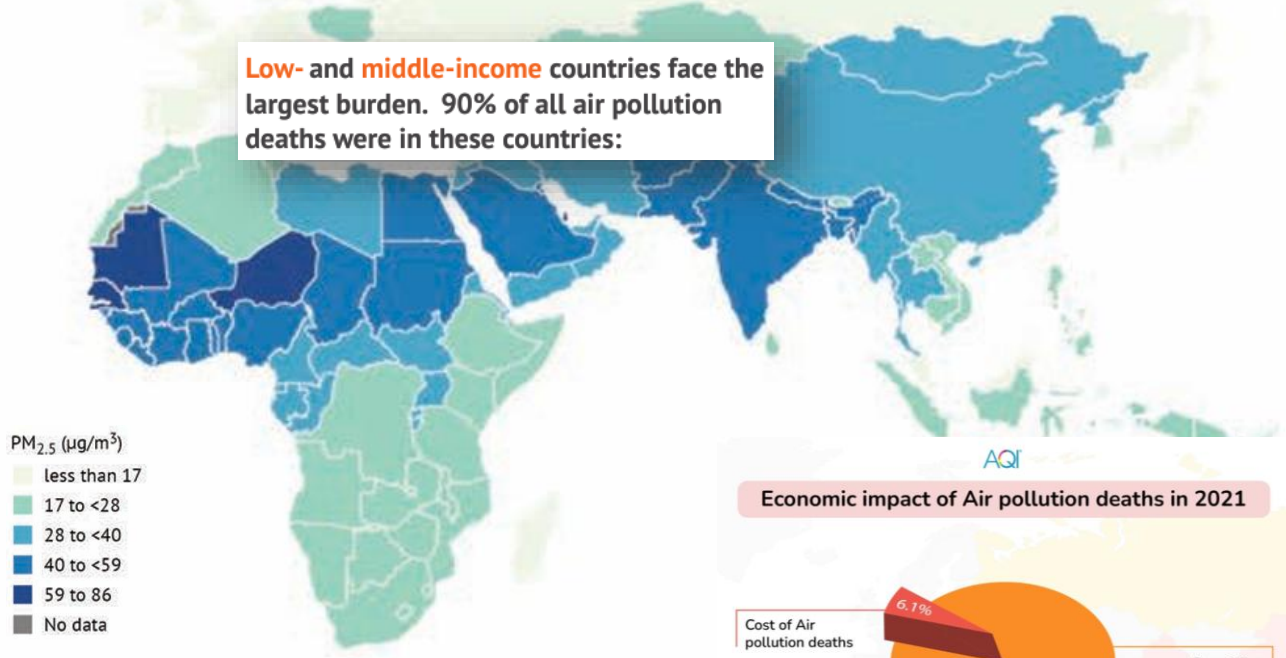
Air pollution, a top global health risk!!



[State of Global Air Report 2025], [State of Global Air Report 2024]

36% of the world's population is exposed to levels of $PM_{2.5}$ above the least stringent interim target of 35 μg/m³ (IT-1).

11% of the global population lives in areas where there are no national air quality standards.

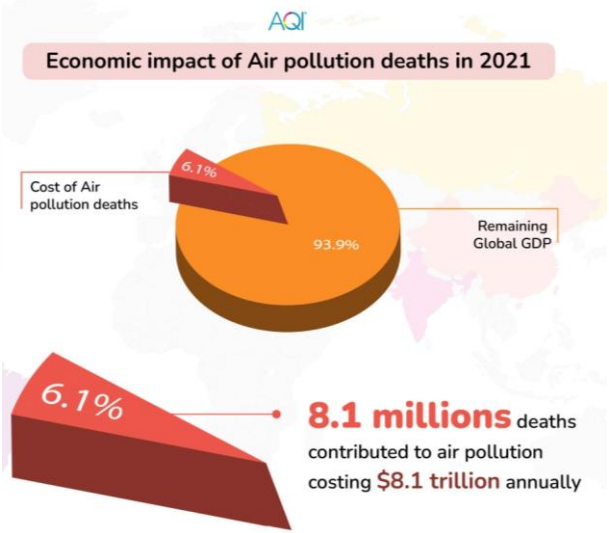


Air pollution continues to be the second leading risk factor for early death, surpassed only by high blood pressure.

7.9 million

deaths attributed to air pollution in 2023,

about **1 in 8** deaths worldwide.



Rank	City
1	Byrnihat, India
2	Delhi, India
3	Karaganda, Kazakhs...
4	Mullanpur, India
5	Lahore, Pakistan
6	Faridabad, India
7	Dera Ismail Khan, P...
8	N'Djamena, Chad
9	Loni, India
10	New Delhi, India

[world-most-polluted-cities]

Traditional/Reference-Grade Monitoring Air Quality Monitoring

BAM (Beta Attenuation Monitor)

Tapered Element Oscillating Microbalance

FRM Gravimetric Samplers

$NO/NO_2/NOx$: Chemi-luminescence

CO_2: NDIR

BTEX, VOCs: Gas chromatography / FID

Strengths

- High accuracy
- Reliable for health decisions
- Provides official data

Limitations

- High cost
- High power demand
- Sparse spatial coverage

The Global Air Quality Monitoring Gap:

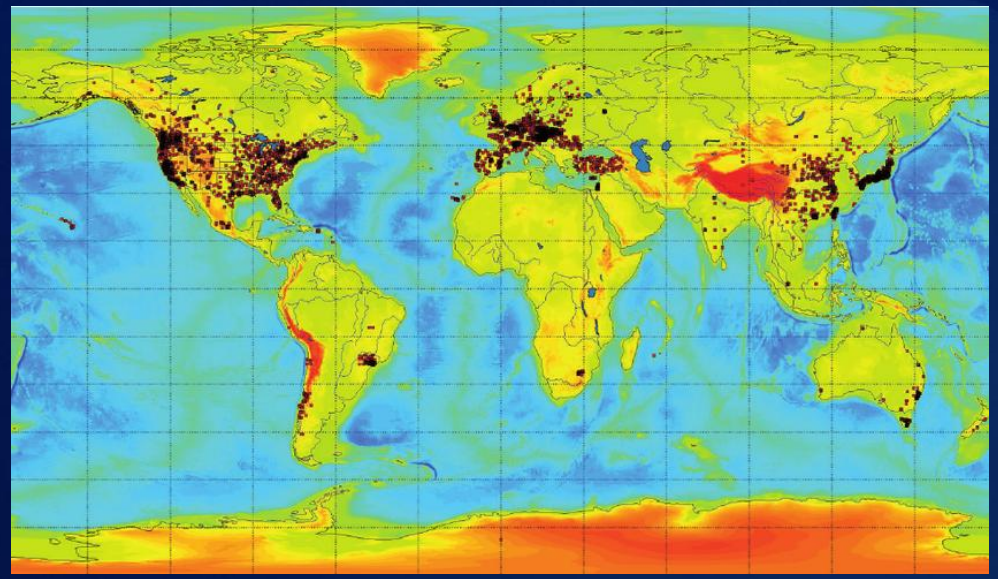
A Deep Dive into the Data Void!!

No. of AQMS in Diff. Continents

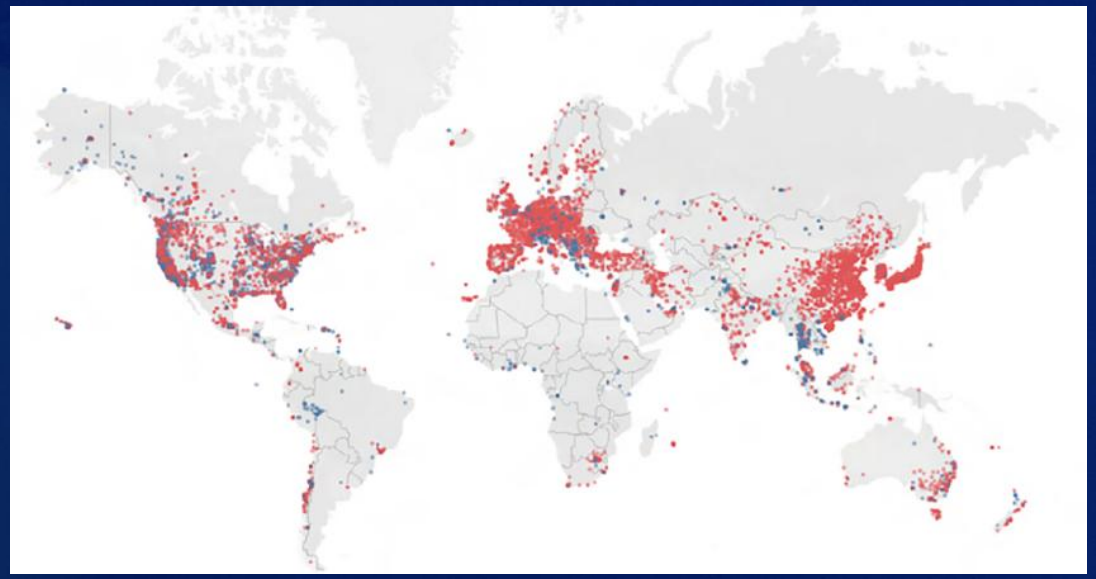
Graphics by macrovector / Freepik

The Global Air Quality Monitoring Gap:

A Deep Dive into the Data Void!!



PM2.5 observation: 8,329 sites in 55 countries (shown as red squares) [Lary et al., 2014]



Global distribution of AQMS. Red dots: Govt. stations; Blue dots: Indep. monitors [E. M. Saleh et al., 2024]

1 per 1M-500k-250k people

1 per 500k-200k-100k people

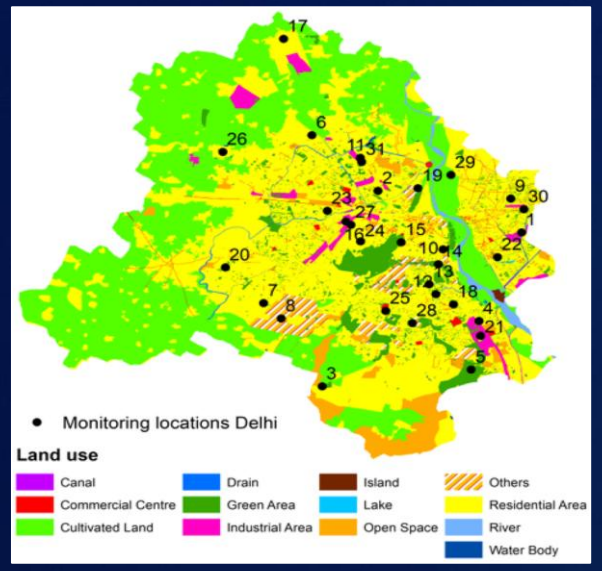
1 per 200-500k people

India's Air Quality Monitoring Gap

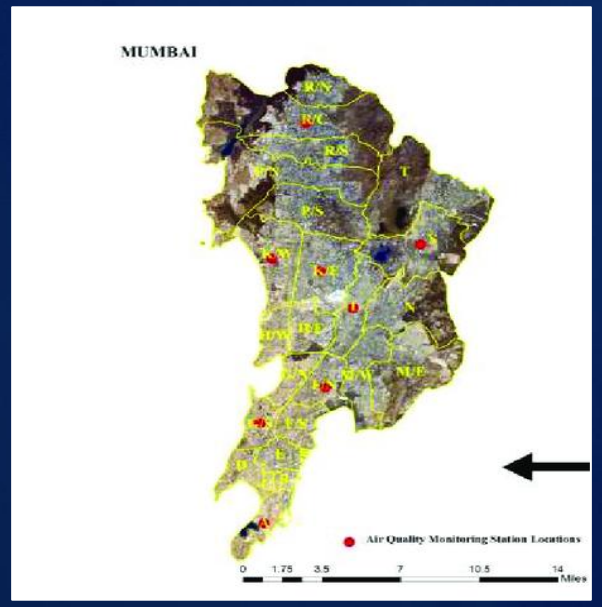
Invisible Crisis, Unmeasured Health Impacts

Programme	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
NAMP	703	731	793	818	882	910	962	965
CAAQMS	114	164	228	279	313	456	538	545

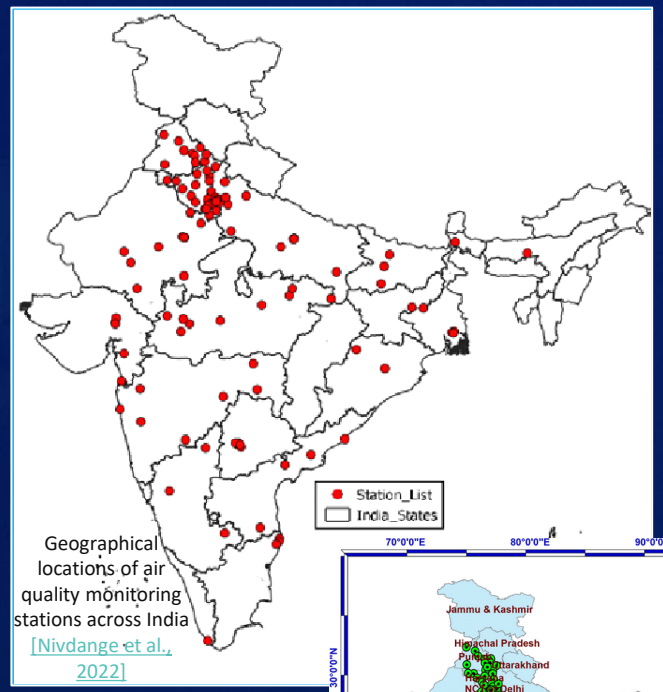
[CPCB: Brief Note On Central Sector Scheme: 'Control Of Pollution']



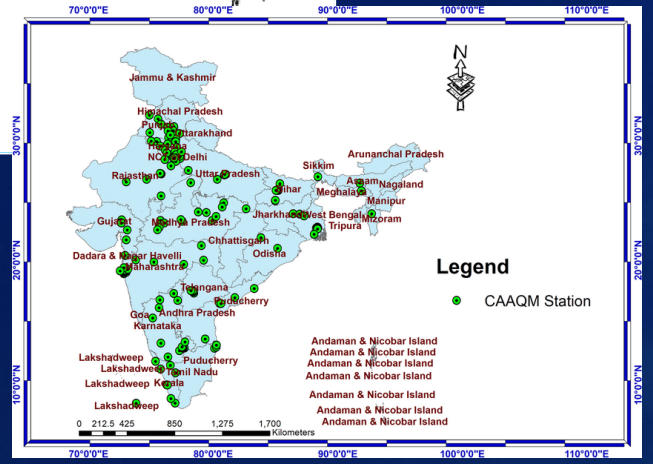
Location of ambient air quality monitoring stations and land use in Delhi [Rahman et al., 2022]



Air quality monitoring station locations on a ward-level map of Mumbai [Nair et al., 2024]



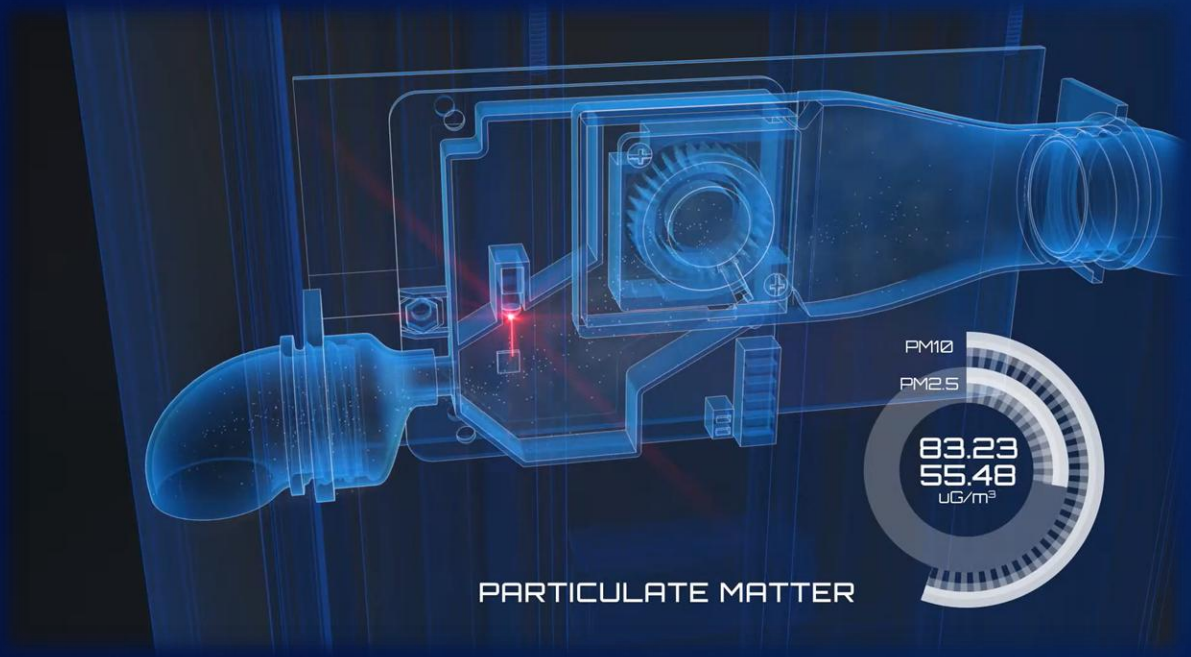
Geographical locations of air quality monitoring stations across India [Nivdange et al., 2022]



Continuous ambient air quality monitoring (CAAQM) stations in India [Periyasami et al., 2022]

Bridging the Air Quality Data Gap
The Imperative for Low-Cost Sensors

Low-cost sensors are measurement devices designed to collect data (such as air quality, temperature, humidity, or pressure) at a **significantly lower price** than traditional, research-grade or industrial sensors.



YT/Oizom

AI Generated

Bridging the Air Quality Data Gap

The Imperative for Low-Cost Sensors

High-Density Networks

- Hyperlocal
- High-res poll. mapping
- Micro-hotspots

Community Empowerment:

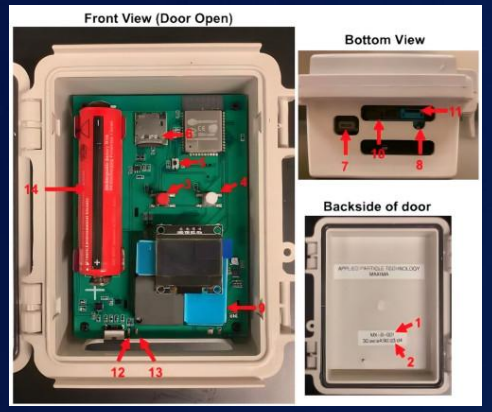
- Democratizes AQ data,
- Public awareness,
- Decision-making.

Supplementary Data:

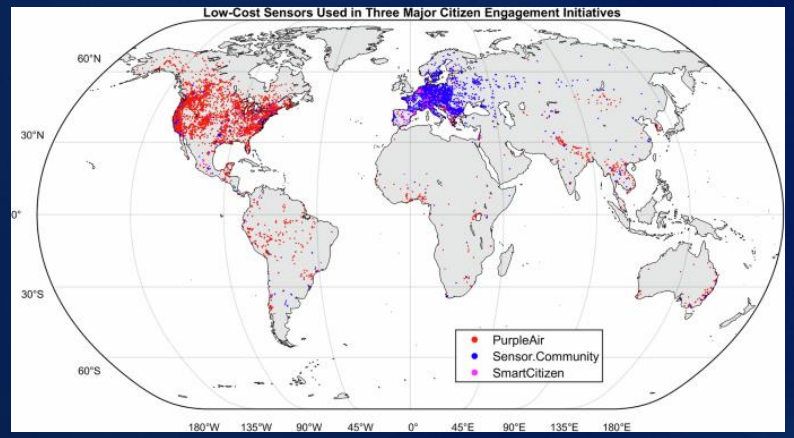
- For model validation,
- Trend analysis,
- Policy input

2020: 40 sensors by IITK & MPCB [HT]

2021, Indore: 19 LCS under Smart Cities Mission [EOS]



LCS measurements closely match the reference instrument. [Dharaiya et al., 2023], [Hayward et al., 2024]



Spatial distribution of fixed LCS [Bagkis et al., 2025]

sameer India's Air Quality Index

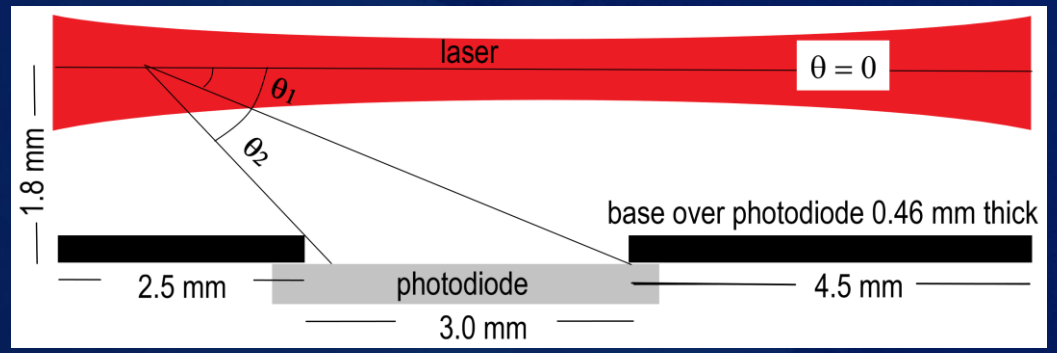
PurpleAir

The Hidden Pitfalls of Low-Cost PM Sensors

Ouimette et al., 2022

PurpleAir PA-PMS

- Inlet losses
- Restricted angular range for laser
- Poor electronic processing
- Variable sensor offsets
- Limited QA/QC
- Hygroscopic growth effects uncertain



PMS internal geometry [Ouimette et al., 2022]

- Individual calibration
- Uncharacterized particle errors
- No single calibration factor
- Potential long-term instability

[Liu et al., 2020]

KOALA AQM

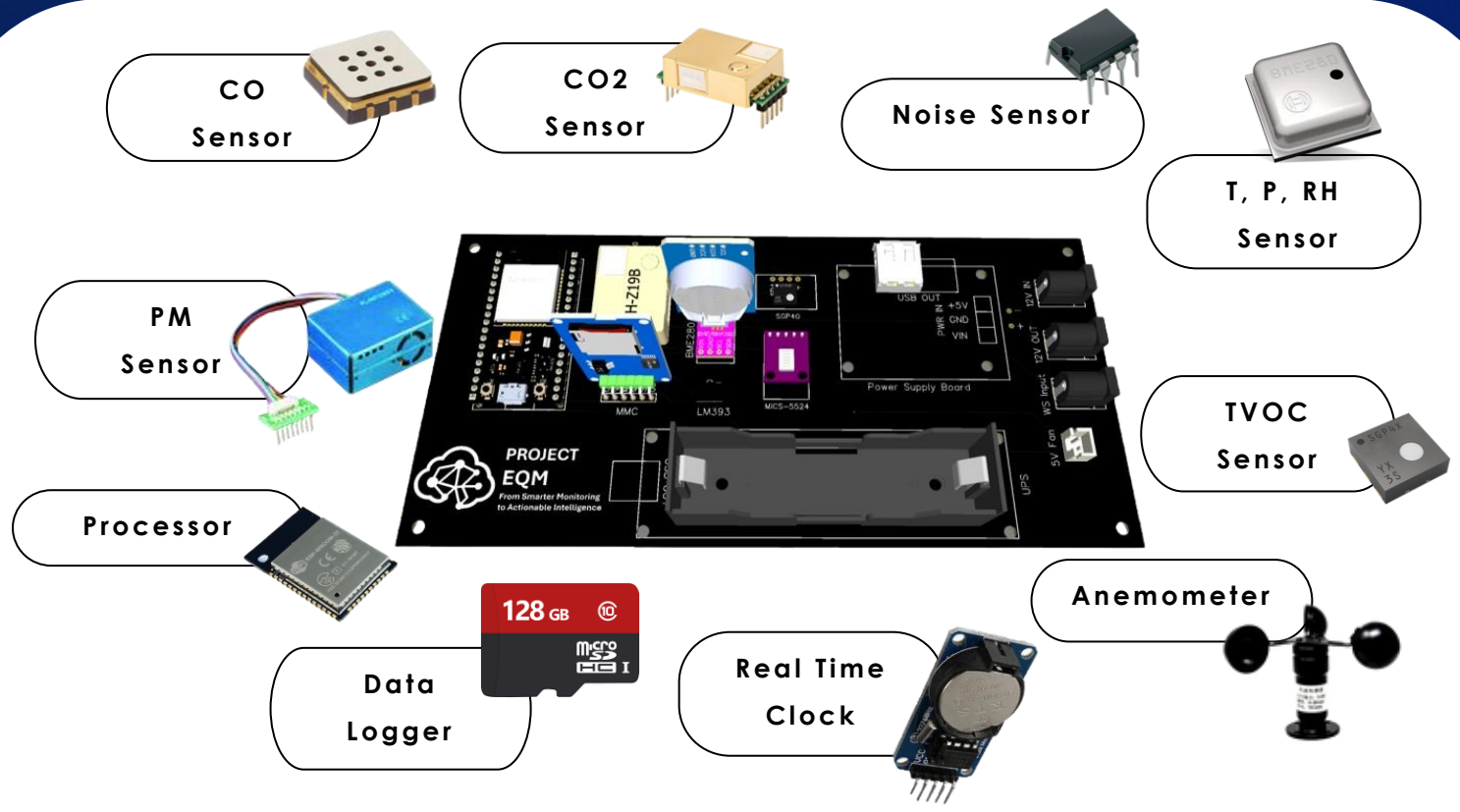
Ouimette et al., 2023

PMS5003

- Flow rate-sensitive
- Does not detect 0.20 µm particles.
- Refractive index-sensitive
- RH-sensitive
- 99% particles miss laser
- Wind speed-sensitive
- Coarse Particle Loss Mechanisms

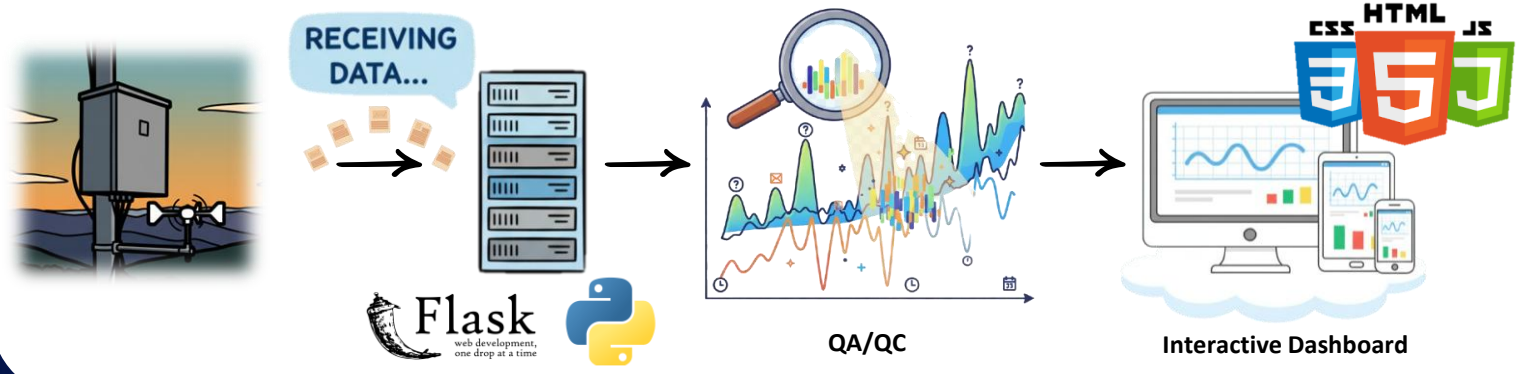
Develop and Deploy an Integrated Low-Cost Environmental Quality Monitor

- Design and deployment of an **integrated low-cost sensor network** for comprehensive monitoring of PM, CO, CO_2, noise, temperature, humidity, wind, and TVOCs.
- Facilitates **city-wide, real-time environmental data acquisition** through a centralized server, enabling seamless monitoring across urban landscapes.

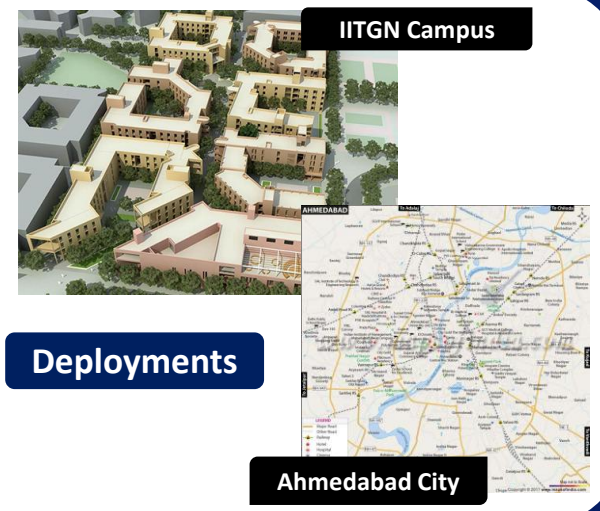


Different components of EQM

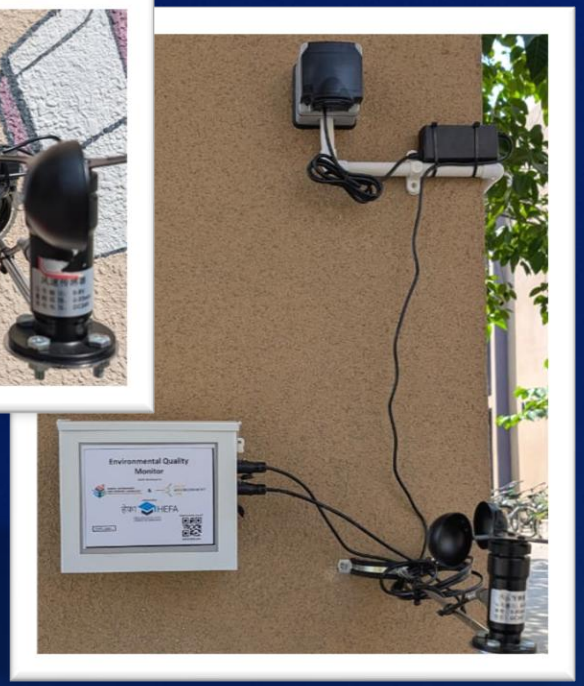
Network Architecture



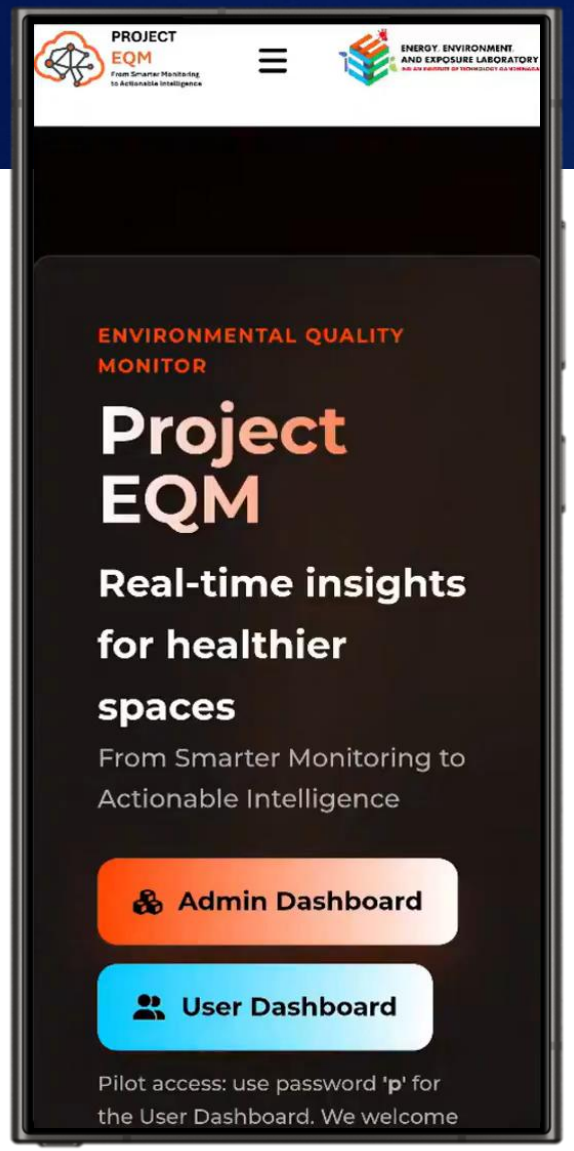
Deployments

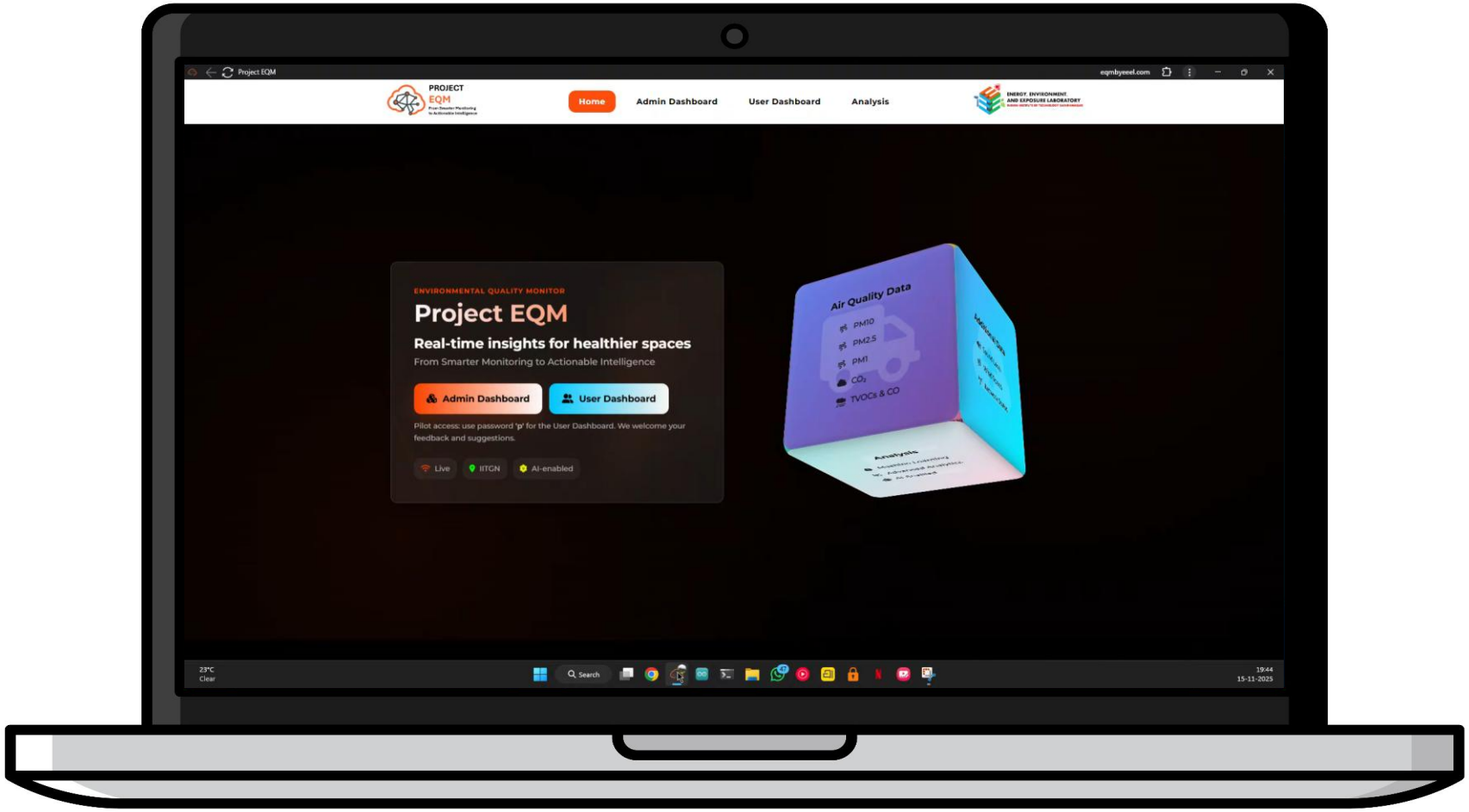


- Initial trial deployment: IITGN campus
- Planned: Large-scale deployment of 50–60 EQM units across Ahmedabad for city-wide, real-time monitoring.
- 1-year continuous monitoring
- Real-time data upload

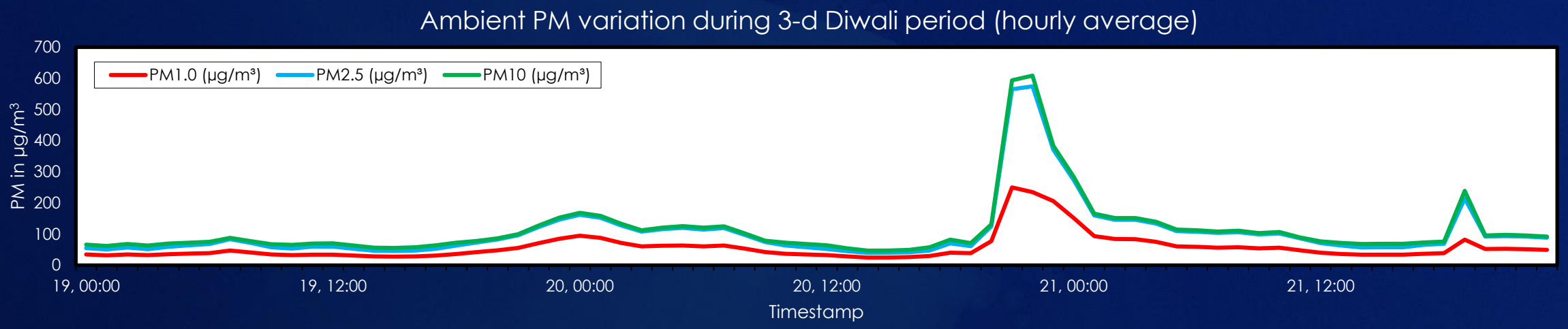


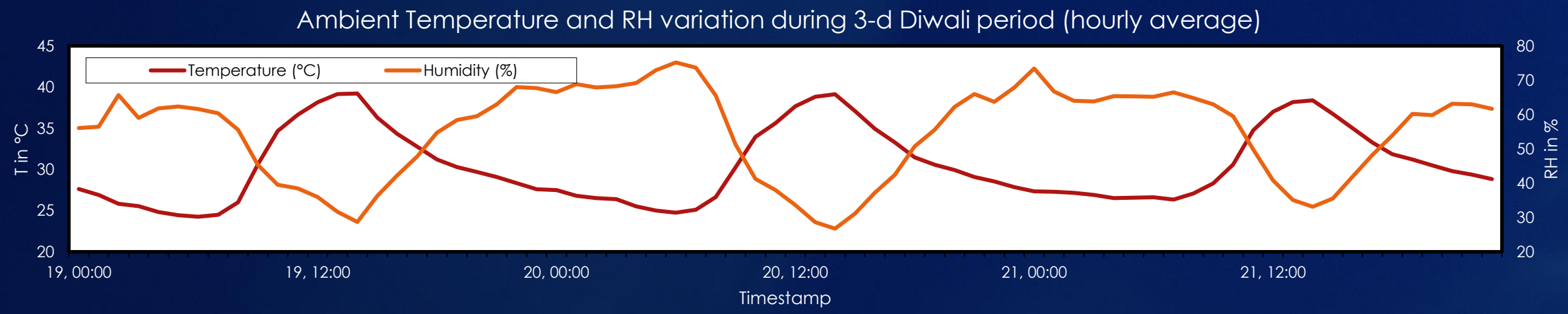
Dashboard


PROJECT
EQM
From Smarter Monitoring to Actionable Intelligence
ENERGY, ENVIRONMENT, AND EXPOSURE LABORATORY
ENVIRONMENTAL QUALITY MONITOR
Project EQM
Real-time insights for healthier spaces
From Smarter Monitoring to Actionable Intelligence
Admin Dashboard
User Dashboard
Pilot access: use password 'p' for the User Dashboard. We welcome


Project EQM
Home
Admin Dashboard
User Dashboard
Analysis
ENVIRONMENTAL QUALITY MONITOR
Project EQM
Real-time insights for healthier spaces
From Smarter Monitoring to Actionable Intelligence
Admin Dashboard
User Dashboard
Pilot access: use password 'p' for the User Dashboard. We welcome your feedback and suggestions.
Live
IITGN
AI-enabled
Air Quality Data
PM10
PM2.5
PM1
CO₂
TVOCs & CO
Analysis

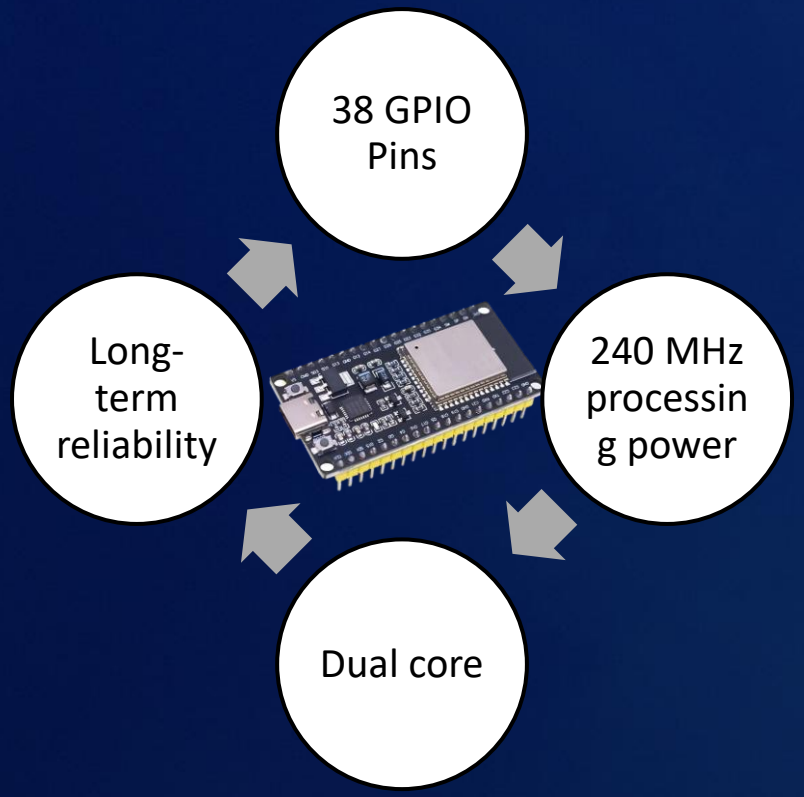
Data Collected From Trial Deployment





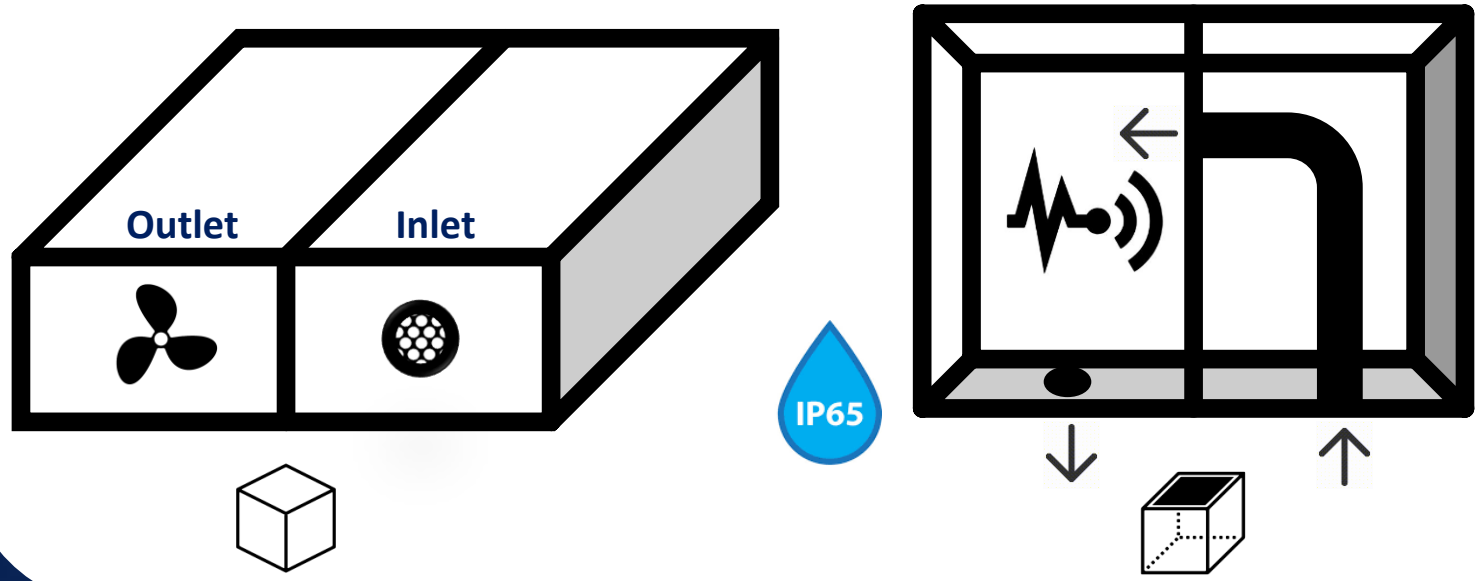
Reliability of the Device

Hardware



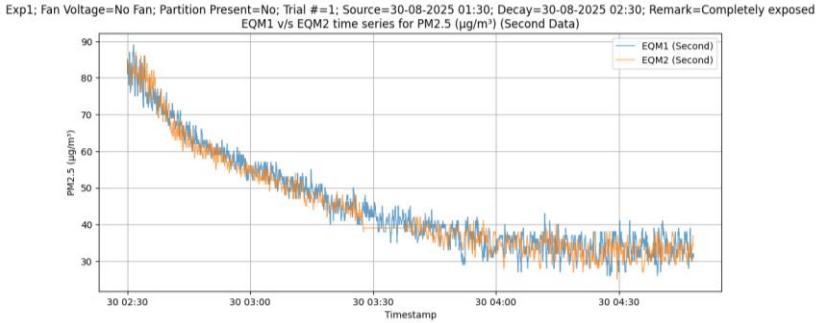
Data Handling

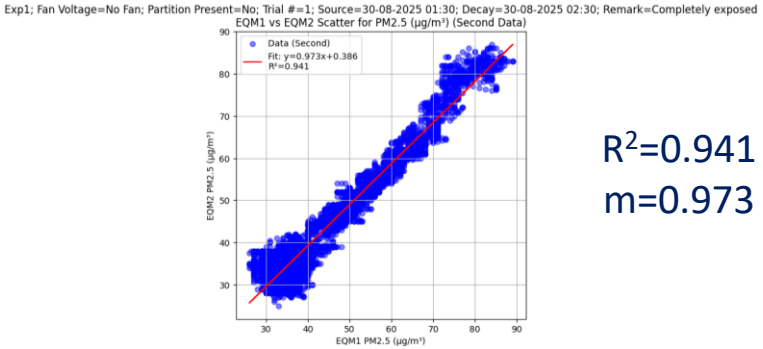
Geometry of Device



A robust, self-reliable system with dependable hardware, fault-tolerant data management, and validated sensing performance.

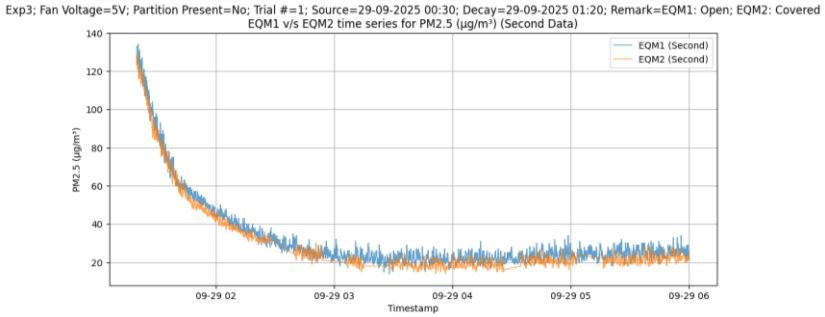
…Reliability of the Device

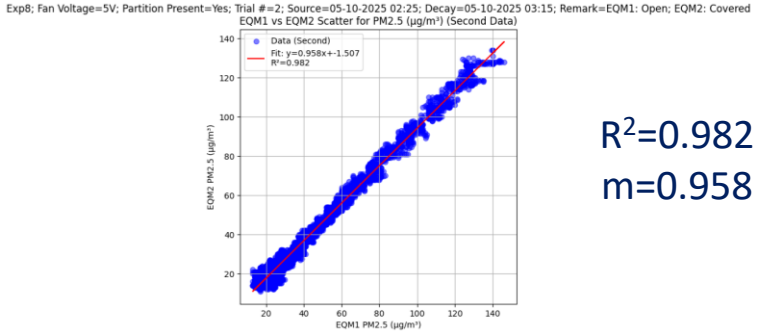




Varying ventilation inside the EQM housing: 0-62-94-226 lpm

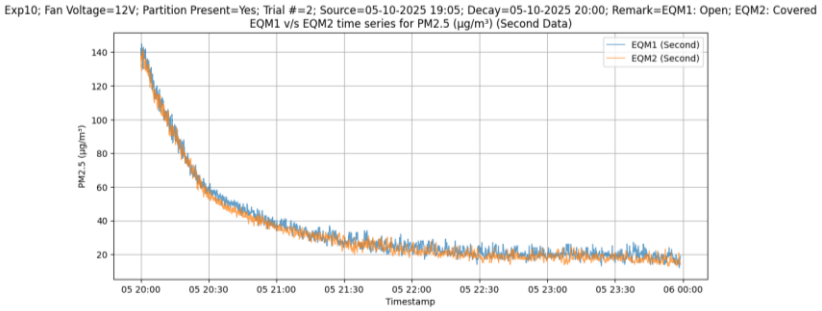
R^2=0.941
m=0.973

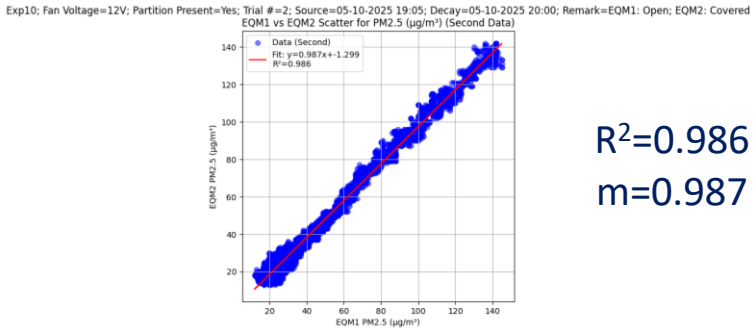




Custom air channelling

R^2=0.982
m=0.958



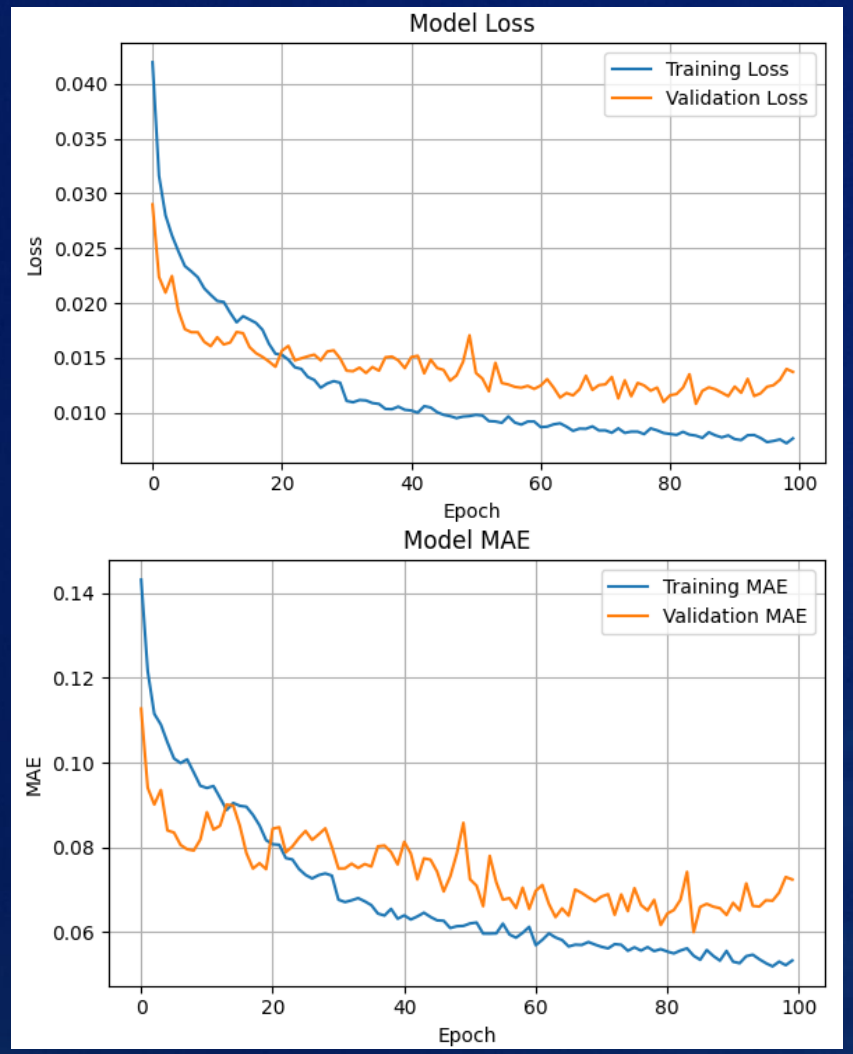


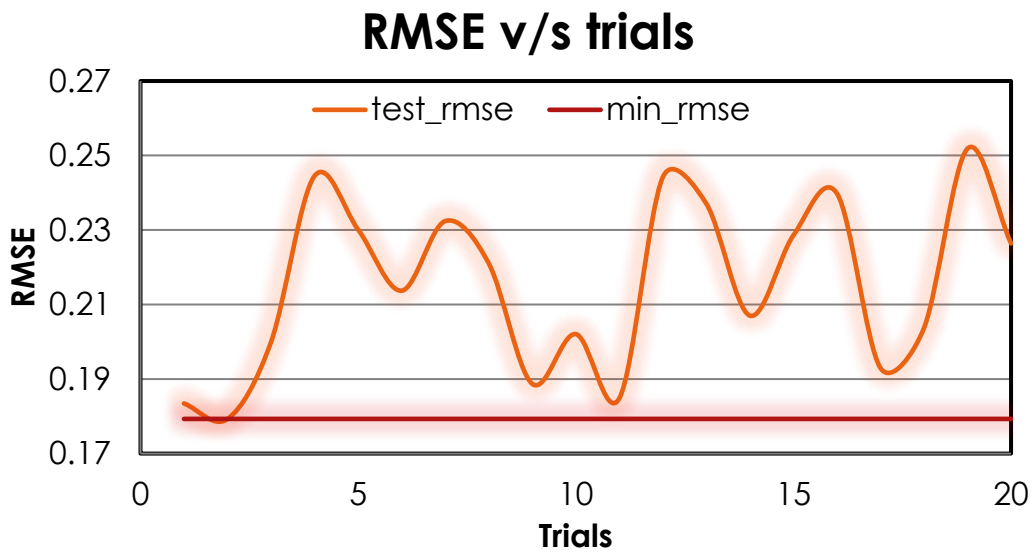
With Proper ventilation & air channels

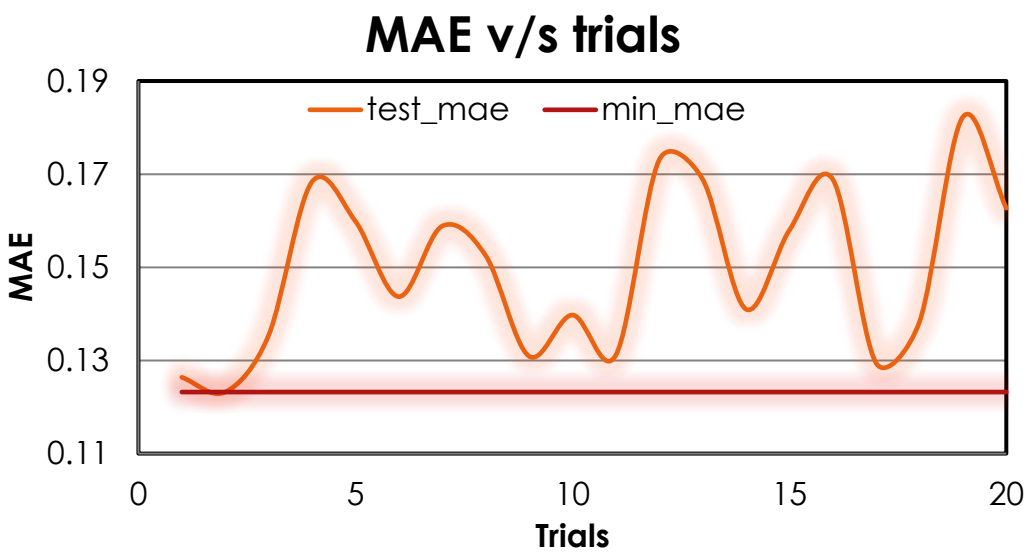
R^2=0.986
m=0.987

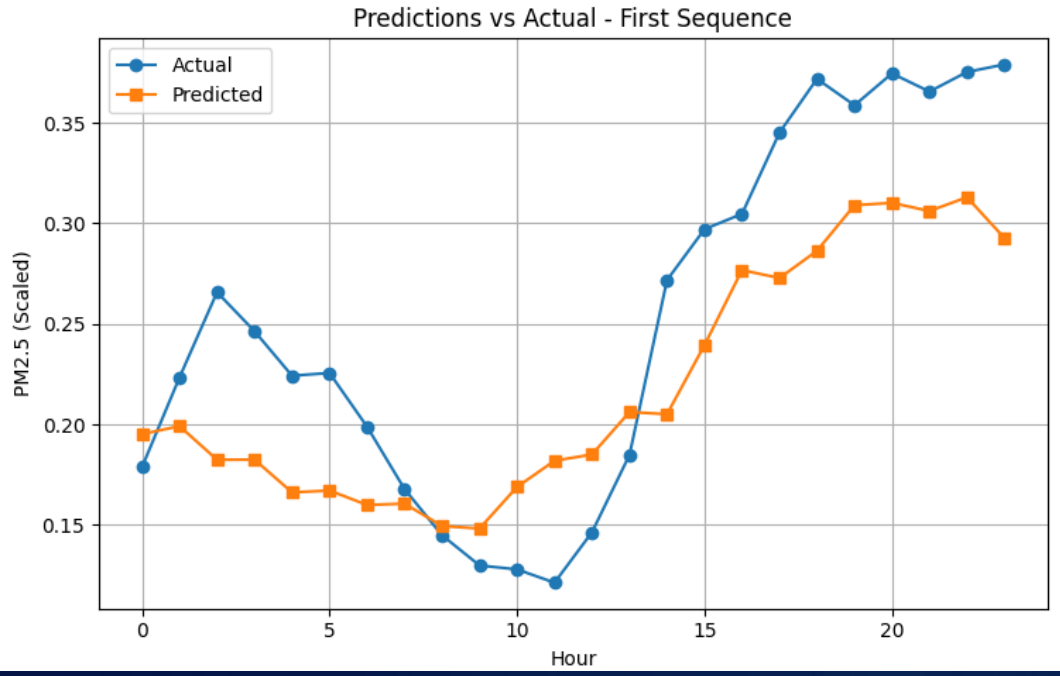
Hybrid CNN-BiLSTM Model for $PM_{2.5}$ Prediction

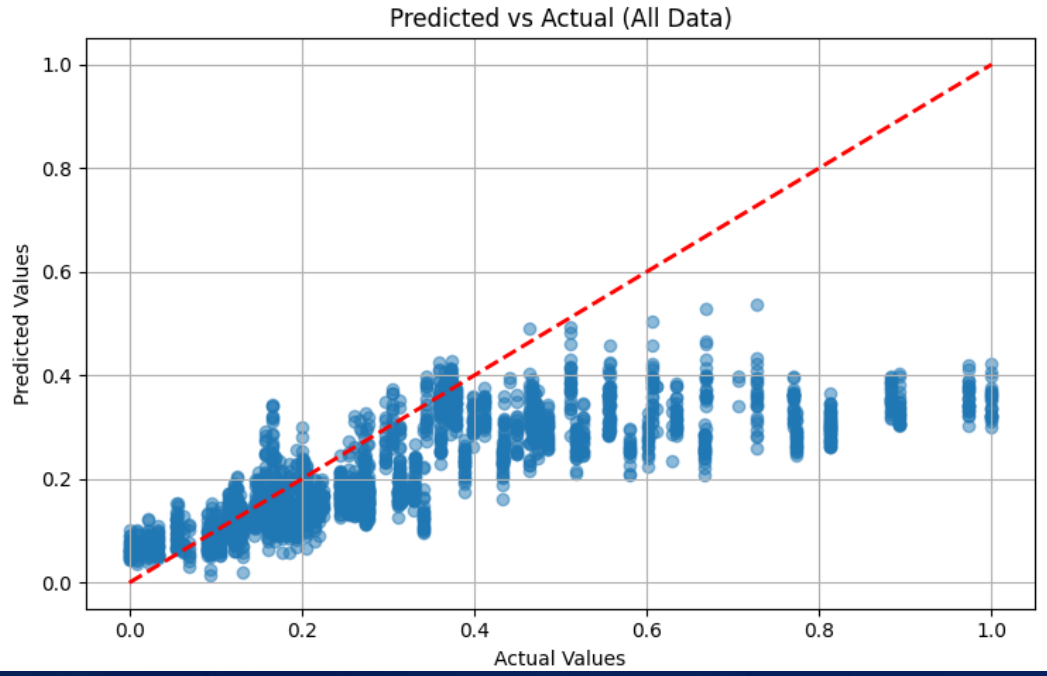
- **Core Architecture: CNN + Bidirectional LSTM**
- **Input:** (PM_1, $PM_{2.5}$, PM_{10}, Temperature, Humidity), (EQM5, EQM6, EQM7, EQM8), and **Min-Max Scaling**
- **1D-CNN Layers: feature extractor** from the time series data, **MaxPooling** layers reduce complexity and highlight the most important features.
- **Bidirectional LSTM Layers: Bi-directionality** allows the model to learn from both past and future context within the sequence, capturing more complex temporal relationships.
- **Output Layer:** A dense layer that outputs a vector of **24 values**, representing the predicted PM2.5 levels for each of the next 24 hours.
- **Optimization:** Used the **Adam** optimizer and **Early Stopping** to prevent overfitting and ensure optimal training.




RMSE v/s trials
test_rmse
min_rmse
RMSE
0.27
0.25
0.23
0.21
0.19
0.17
0
5
10
15
20
Trials


MAE v/s trials
test_mae
min_mae
MAE
0.19
0.17
0.15
0.13
0.11
0
5
10
15
20
Trials


Predictions vs Actual - First Sequence
Actual
Predicted
PM2.5 (Scaled)
0.35
0.30
0.25
0.20
0.15
0
5
10
15
20
Hour


Predicted vs Actual (All Data)
Predicted Values
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Actual Values

Conclusions & Future Work

1. Major monitoring gaps persist globally, especially in the Global South, and scalable low-cost sensors can significantly expand air-quality coverage.
2. The EQM device combines multiple low-cost sensors to reliably gather environmental data and transmit it to a centralised monitoring server.
3. Strong hardware and data pipelines ensure reliability, though sensor drift and cross-sensitivity require calibration and fusion for accurate performance.
4. The Hybrid CNN-BiLSTM $PM_{2.5}$ model performs poorly at high concentrations but can improve with more data, tuning, and added meteorology.
5. Future work includes adaptive calibration, redundant sensing, large-scale deployment, and advanced models using richer datasets for better spatiotemporal predictions.

Acknowledgement

Prof. S. Patel

Prof. S. Sen

A. K. Thakur

N. K. Mishra

C. Nagar

हेफ़ा HEFA
ENABLING EXCELLENCE
उत्कृष्टता को सक्षम कराना

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Thank you for your attention.
Questions, comments, or insights?